HEALTHCARE INNOVATION & SYNCHRONIZATION PLATFORM

MediSync System

Secure Hospital-to-Pharmacy Synchronization System

PRESENTER
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ROLE
Full-Stack Developer

INSTITUTION
[Institution / Company Name]

EXECUTIVE SUMMARY

Project Overview

What MediSync Is

MediSync is a web-based healthcare coordination platform that connects hospitals, doctors, patients, and pharmacies through a centralized digital prescription workflow.

Core Value Drivers

- Digital prescription transmission from doctors to verified pharmacies.
- Digital prescription verification using QR-based validation.
- Centralized patient medical history and PDF export.
- System audit logging for operational transparency and governance.

HEALTHCARE GAPS

Problem Statement

GAP 01

Paper Prescription Risks

Traditional handwritten and printed prescriptions can increase the risk of alteration, duplication, and dispensing errors.

GAP 02

Fragmented Pharmacy Workflows

Manual verification between pharmacies and prescribing providers can introduce delays and operational bottlenecks.

GAP 03

Fragmented Patient Records

Patients may lack convenient access to centralized digital records of prescriptions, medications, and prescribing information.

THE INNOVATION

Proposed Solution

1. PROBLEM

Manual & Vulnerable

Paper prescription delays, error-prone dispensing, unverified pharmacy interactions.

2. SOLUTION

MediSync Platform

Role-based authentication, digital prescription transmission, QR-based verification, and centralized record management.

3. OUTCOME

Verified Digital Workflow

Fast digital prescription workflow, audit traceability, traceable and verifiable prescription lifecycle.

STRATEGIC ALIGNMENT

Core System Objectives

1. Fast Digital Prescription Sync

Deliver prescriptions to verified pharmacies through a fast digital workflow.

2. Role-Based Access Control

Enforce role-based access controls across Admin, Hospital, Doctor, Pharmacy, and Patient portals.

3. Application Security

Implement JWT-based authentication, password hashing, rate limiting, and request validation.

4. Verification Governance

Implement mandatory Admin approval queues for new Hospital and Pharmacy registrants.

5. Audit Logging

Maintain system audit trails for prescription generation, link requests, and dispensing events.

6. Digital Health PDF Exports

Generate downloadable, standardized prescription documents for offline access and record keeping.

USER ECOLOGY

Multi-Role User Personas

ROLE 1

System Admin

Oversees platform governance, approves hospital & pharmacy accounts, monitors system audit logs.

ROLE 2

Doctor

Creates digital prescriptions, links patient records, assigns consulting rooms and medical instructions.

ROLE 3

Pharmacy Staff

Accesses pending prescription queues, verifies dosage details, and marks prescriptions as DISPENSED.

ROLE 4

Patient

Views assigned doctor details, tracks prescription status, downloads PDF records, scans QR verification.

CAPABILITIES

Implemented Core Features

Digital Prescription Engine
Dynamic medicine addition with dosage, frequency, and duration tracking.

Near-Real-Time Status Synchronization
Prescription status changes are reflected across relevant views through API synchronization.

QR-Based Verification & PDF Generation
Built-in QR code generator and client-side jsPDF document compilation.

Hospital-Patient Linking
Patient linkage mechanism establishing provider-patient care relationship.

Admin Approval System
Account authorization requirement preventing unverified entities from issuing medications.

Audit Trail & System Monitoring
Automated system audit logs recording action type, target user, timestamp, and metadata.

PROCESS FLOW

End-to-End System Workflow

STEP 1

Consultation

Doctor creates digital prescription in Doctor Portal.

STEP 2

API Processing

Express API validates the request, stores the prescription, and records the corresponding audit event.

STEP 3

Pharmacy Queue

Prescription appears in near-real-time on Pharmacy Queue.

STEP 4

Fulfillment

Pharmacy dispenses medication; status updates to DISPENSED across relevant views.

ENGINEERING

Technology Stack

Frontend Tier

- React 19 & Vite 7
- Tailwind CSS v4
- Framer Motion 12
- Lucide React & jsPDF 4

Backend Tier

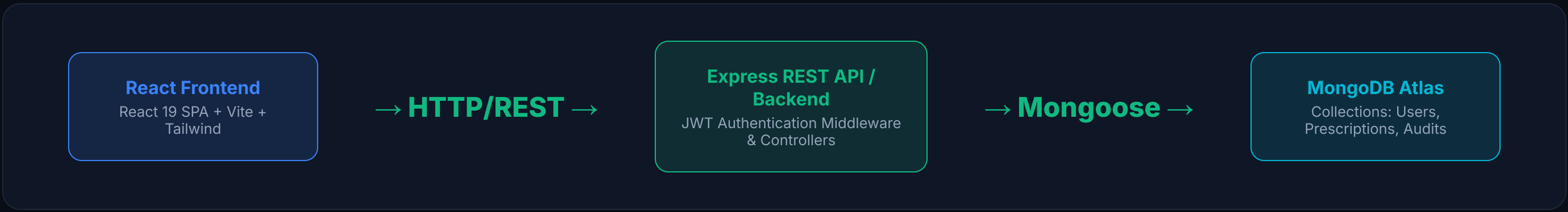
- Node.js & Express 5
- JWT & Cookie Parser
- BcryptJS 3 Password Hashing
- Express Rate Limit 8

Database & Services

- MongoDB Atlas (Mongoose 9)
- Nodemailer 8 Notification Service
- Axios HTTP Client
- QRCode React 4 Component

SYSTEM TOPOLOGY

System Architecture



DATA ARCHITECTURE

Data Schema & Entity Relationships

User Schema

email, passwordHash, role, isApproved, hospitalDetails (licenseNumber, roomNumber, floorNumber), assignedDoctor.

Prescription Schema

patientId, patientName, doctorId, doctorEmail, medicines [name, dosage, frequency, duration], status (pending/dispensed).

AuditLog Schema

action, performedBy, targetUser, description, timestamp.

Entity Relationships

User (Doctor / Patient / Pharmacy) → Prescription → Medicine | User → AuditLog

VISUAL VERIFICATION

Doctor & Hospital Interface



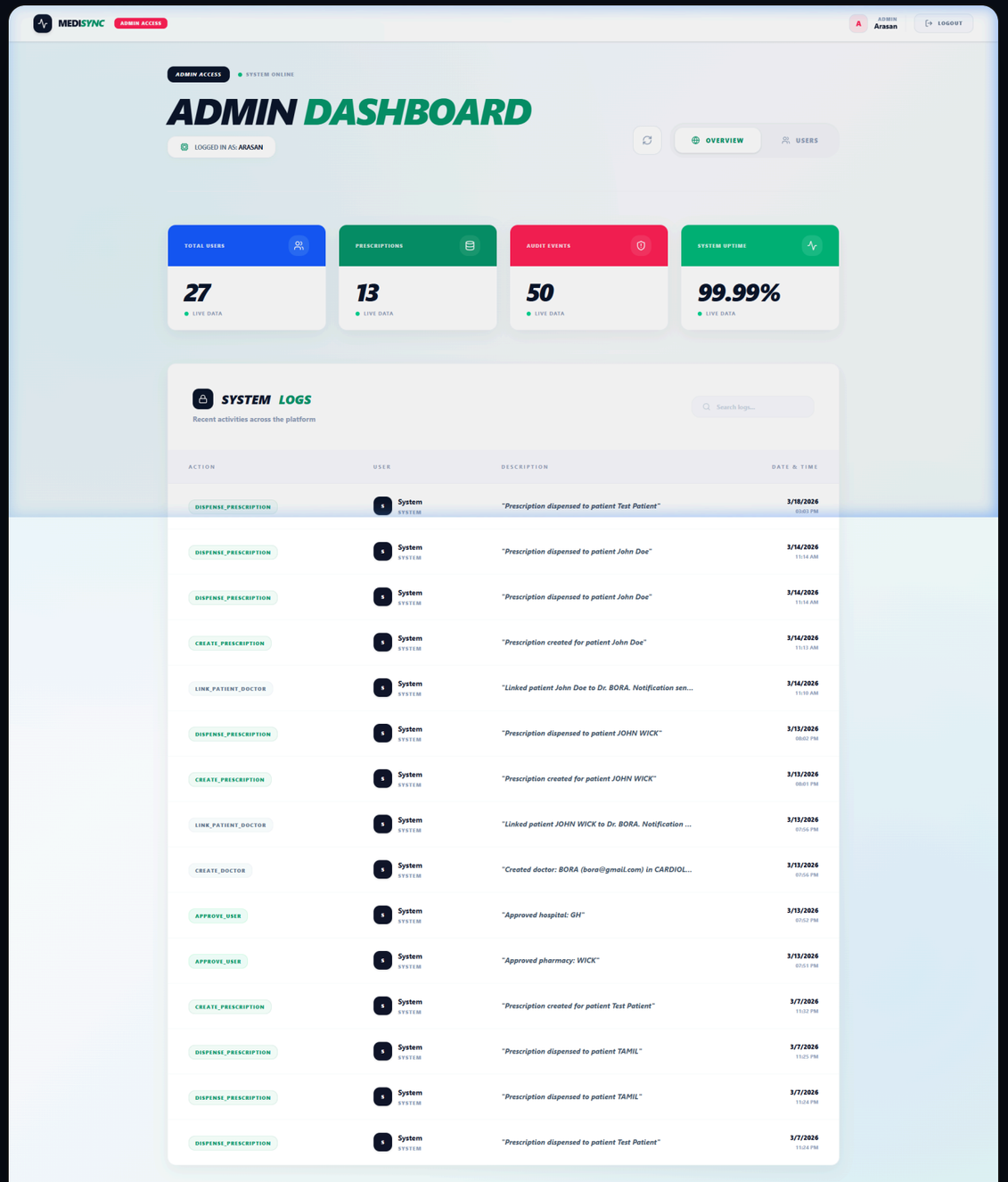
Doctor Portal Highlights

- Prescription creation modal with dynamic medication lists.
- Metrics panel tracking total issued vs. pending prescriptions.
- Connection and system status indicators.
- Direct room assignment and hospital communication status.

Admin Dashboard & Operations

Platform Control & Governance

- User verification queue for newly registered hospitals and pharmacies.
- Platform Audit Log displaying system actions with user attribution.
- Platform health indicators and operational metrics.
- Global entity management and permission revocation controls.



DATA PROTECTION

Application Security Measures

HTTP-Only Cookie Auth

HTTP-only cookies reduce client-side JavaScript access to authentication tokens and help mitigate token theft through certain XSS scenarios.

BcryptJS Password Hashing

Passwords are hashed using bcryptjs before storage in the database.

Rate Limiting & Validation

Rate limiting helps protect authentication endpoints against repeated or brute-force login attempts.

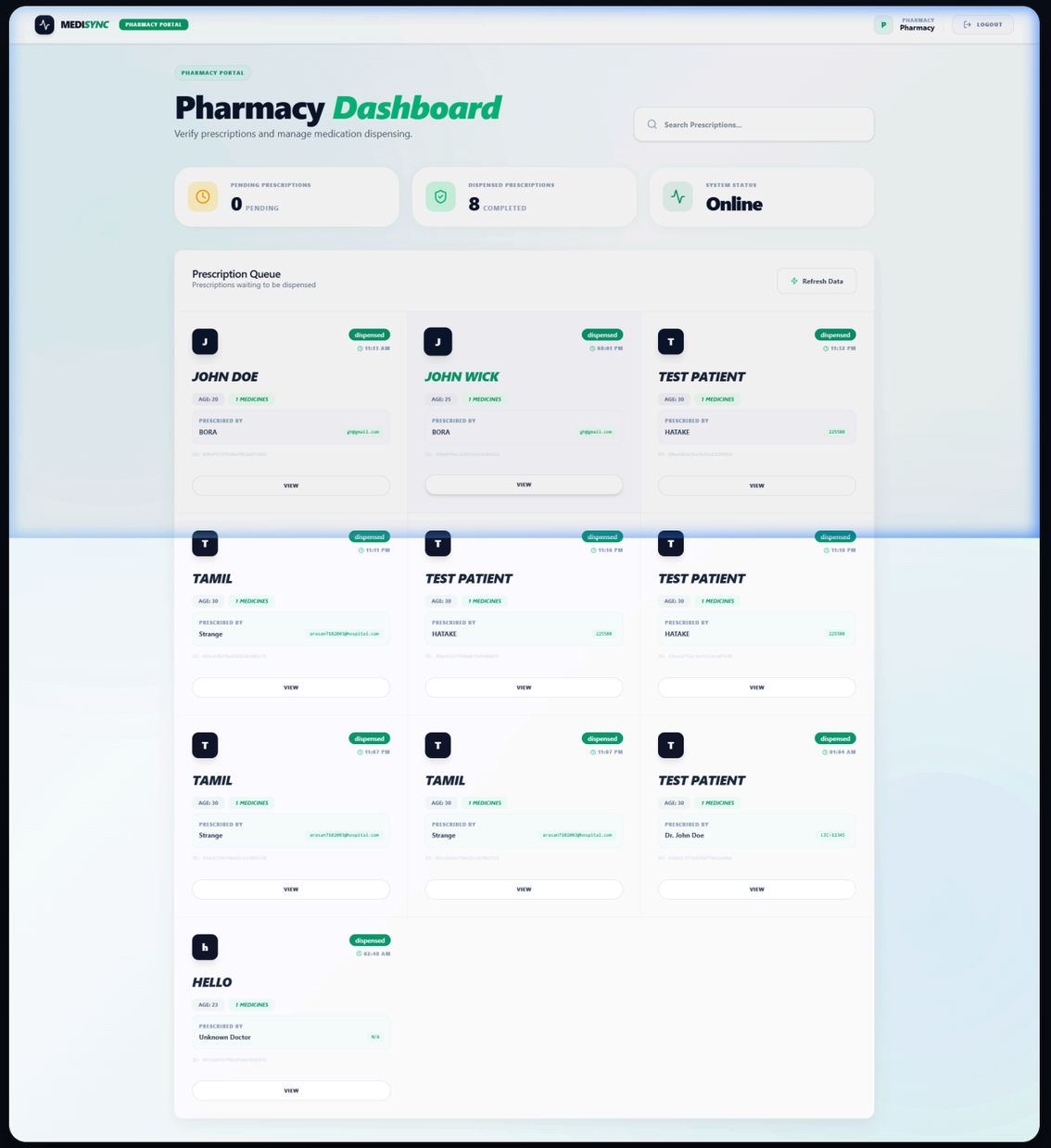
QUALITY ASSURANCE

Functional / Manual Testing

Test Module	Scope / Expected Result	Actual Result	Status
Authentication	Correct role-based access	Verified	PASS
Prescription Creation	Prescription stored successfully	Verified	PASS
Dispensing Workflow	Status changes correctly	Verified	PASS
PDF Export	PDF generated successfully	Verified	PASS

DEPLOYMENT STATUS

Results & Platform Status



Verified Achievements

- Functional multi-role platform (Admin, Doctor, Pharmacy, Patient).
- Near-real-time prescription queue synchronization between doctors and pharmacies.
- Verified database persistence with MongoDB Atlas cloud connectivity.
- End-to-end dispensing workflow completed successfully during functional testing.

ENGINEERING INSIGHTS

Technical Challenges & Solutions

CHALLENGE 1

Cross-Role State Sync

Problem: Ensuring prescriptions created by doctors update in pharmacy queues without full manual page reloads.
Solution: Polling and structured REST API refetching were used to keep prescription status synchronized across relevant views.

CHALLENGE 2

Secure Multi-Role Access Control

Problem: Preventing unauthorized access to patient data across different roles.
Solution: Enforced server-side JWT verification middleware on API endpoints checking user role and account approval state.

PRODUCT ROADMAP (PLANNED SCOPE)

Future Enhancements (Planned)

PLANNED

FHIR / HL7 Interoperability

Standardized data exchanges with legacy hospital electronic health record (EHR) systems.

PLANNED

Native iOS & Android Apps

React Native mobile application for mobile prescription scanning and push notifications.

PLANNED

AI-Assisted Drug Interaction Detection

Future AI-assisted analysis for identifying potential drug-drug interactions using validated clinical data.

SUMMARY

Conclusion & Value Impact

MediSync Platform Impact

MediSync provides a centralized digital workflow for prescription creation, verification, dispensing, and record management, improving traceability and reducing reliance on manual processes.

DIGITIZING PRESCRIPTION WORKFLOWS THROUGH SECURE, CONNECTED TECHNOLOGY

Thank You

Live Product Demo & Q&A Session

Project: **MediSync**
Presenter: **Tamilarasan**
Status: **Deployed Prototype**